

COMPREHENSIVE STUDY GUIDE

Physical Pharmaceutical Chemistry

PCH 201 · PHC 201 — Foundations, Solutions, Drugs, Ionic Equilibria,
Thermodynamics & Reaction Kinetics

PART ONE · CHEMISTRY IN PHARMACY

Roles of chemistry · molecular representation · stereochemistry

PART TWO · DRUGS IN THE LIQUID PHASE

Systems · dispersions · solutions · colligative properties · electrolytes

PART THREE · BEHAVIOUR OF DRUGS

Definitions · discovery & design · sources · routes · ADME & pharmacodynamics

PART FOUR · IONIC EQUILIBRIA

Equilibrium constant · pH · K_a/K_b & pK_a · Henderson-Hasselbalch · buffers

PART FIVE · THERMODYNAMICS

Laws · enthalpy · entropy · Hess's law · Gibbs energy · drug-receptor binding

PART SIX · CHEMICAL KINETICS

Rate laws & order · half-life & shelf-life · factors · enzyme kinetics

An original study guide by the EverythingABUAD team.

Table of Contents

PART ONE — CHEMISTRY IN PHARMACY

1. The Roles of Chemistry in Pharmacy

Analysis, synthesis, computation, natural products, therapy

2. Representing Molecules: 1D, 2D & 3D

Why the third dimension matters for drug-receptor fit

3. Natural Products, Stereochemistry & the Pharmacist

Enantiomers, stereocentres and the pharmacist's edge

PART TWO — DRUGS IN THE LIQUID PHASE

4. Pharmaceutical Systems & Dispersions

Systems, phases, molecular / colloidal / coarse dispersions

5. True Solutions: Solvent, Solute & Types

Binary solutions and the nine solute-solvent combinations

6. Physical Properties of Substances

Colligative, additive, constitutive · extensive vs intensive

7. Electrolytes, Ideal Gases & Ideal Solutions

Strong, weak and non-electrolytes; ideal behaviour

PART THREE — BEHAVIOUR OF DRUGS IN BIOLOGICAL SYSTEMS

8. What Are Drugs? Definition & the Need for New Drugs

WHO definition, side effects, drivers of drug development

9. Drug Discovery & Design: A Historical Outline

Ehrlich, SAR/QSAR, receptors, ligands and pharmacophores

10. Sources of Drugs

Natural, synthetic, semi-synthetic and biosynthetic origins

11. Routes of Administration & Dosage Forms

Enteral vs parenteral · factors governing route choice

12. Pharmacokinetics (ADME) & Pharmacodynamics

Absorption-distribution-metabolism-excretion; PK/PD integration

PART FOUR — IONIC EQUILIBRIA

13. Chemical Equilibrium & the Equilibrium Constant

Dynamic equilibrium and the derivation of K

14. Water Equilibrium & the pH Scale

K_w, the ionic product of water, pH and pOH

15. Dissociation of Weak Acids & Bases

K_a, K_b, pK_a and comparing acid strengths

16. The Henderson-Hasselbalch Equation

Deriving and applying the buffer equation

17. Buffer Solutions: Mechanism & Types

How buffers resist pH change; acidic and basic buffers

18. Buffers in Pharmacy & Biological Systems

Formulation roles and the body's own buffer systems

19. Worked Examples & Tutorial Problems

Fully solved pH and buffer calculations for revision

PART FIVE — THERMODYNAMICS

20. Foundations of Thermodynamics

Systems, boundaries, processes & state functions

21. The Zeroth Law & Thermal Equilibrium

Temperature as a transitive property

22. The First Law of Thermodynamics

Energy conservation, enthalpy & reaction types

23. Reversibility, Equilibrium & K

Le Chatelier's principle & the equilibrium constant

24. The Second Law & Entropy

Spontaneity, randomness & the arrow of time

25. Heat of Reaction & Kirchhoff's Equations

Standard states & variation with temperature

26. Types of Heat of Reaction

Formation, combustion, solution, fusion & more

27. Hess's Law of Constant Heat Summation

Enthalpy as a state function · worked examples

28. The Third Law of Thermodynamics

Absolute entropy at 0 K · Nernst theorem

29. Standard & Numerical Entropy

Clausius definition & entropy of reaction

30. Gibbs Free Energy & Spontaneity

$\Delta G = \Delta H - T\Delta S$ · temperature & pressure effects

31. Free Energy, Equilibrium & K

$\Delta G^\circ = -RT \ln K$ · the reaction quotient Q

32. Thermodynamics of Drug-Receptor Binding

Ligand affinity & enthalpy-entropy compensation

PART SIX — CHEMICAL KINETICS**33. Reaction Kinetics & Drug Stability**

Shelf life, stability & stress testing

34. Rate of Reaction, Rate Laws & Order

The law of mass action & pseudo-order

35. First-Order Reactions

Integrated law, half-life & shelf-life

36. Second-Order Reactions

Bimolecular kinetics & the $1/[A]$ plot

37. Zero-Order Reactions

Concentration-independent rate

38. Factors Affecting Reaction Rate

Temperature, concentration, catalysis & more

39. Enzyme Kinetics & Michaelis-Menten

$V_{\max}$, K_m & enzyme-substrate mechanism

PART ONE

CHEMISTRY IN PHARMACY

§1 THE ROLES OF CHEMISTRY IN PHARMACY

Pharmacy is, at its core, an applied chemical science: every medicine a pharmacist handles is a chemical entity, and understanding its behaviour depends on understanding chemistry. The discipline is best appreciated not as a single subject but as a family of overlapping branches, each answering a different professional question about a drug. A useful way to enter the field is to look at how each branch of chemistry contributes to the pharmacist's work.

1.1 The branches of chemistry a pharmacist meets

Branch	Central question it answers	Typical pharmacy application
Analytical chemistry	What is in this product, and how much?	Assay of active ingredient, detection of fake or substandard products, identification and purity testing
Organic / medicinal chemistry	How is the molecule built and how does it react?	Drug design, identification of structures, and understanding reactions that occur in and outside the body
Computational chemistry	What are the molecule's properties before we make it?	Computing physicochemical properties and designing entirely new candidate molecules
Inorganic & physical chemistry	Why does the drug behave this way physically?	Explaining solubility, stability and energetics of drug systems
Biochemistry	How does the drug interact with living tissue?	Linking chemical structure to biological and therapeutic effect

1.2 Analysis, synthesis and therapy

Analytical work underpins quality: it lets a pharmacist confirm the composition of a drug, expose counterfeit or substandard products, quantify the active ingredient through assay, and confirm both identity and purity. Organic and medicinal chemistry supply the creative side of the profession — the design of new drugs, the elucidation of chemical structures, and the study of reactions that molecules undergo either inside the body (in vivo) or in the laboratory (ex vivo). Computational chemistry increasingly sits alongside these, allowing properties to be predicted and molecules to be designed on a computer before any synthesis is attempted.

PHARMACEUTICAL RELEVANCE

It is chiefly this command of chemistry that gives the pharmacist a distinctive edge within the healthcare team. The goal of pharmaceutical education is not to memorise and recite chemical facts in an examination, but to produce practitioners who can resolve real therapeutic problems using chemical reasoning.

§2 REPRESENTING MOLECULES: 1D, 2D AND 3D

Drug molecules are three-dimensional objects, yet they are most often drawn on paper in one or two dimensions. Recognising the limits of these simplified representations is essential, because a drug's activity depends on its real spatial shape.

2.1 Why the third dimension matters

Paracetamol, with the molecular formula $C_8H_9NO_2$, can be written as a flat structural diagram, but its atoms actually occupy positions above, below and within the plane of its benzene ring. Capturing that reality requires a third axis (the z-axis) in addition to the two used on the page. Hydrogen atoms, for instance, may lie in front of the ring plane, behind it, or within it — a distinction a flat drawing cannot convey.

Using a two-dimensional picture to describe a three-dimensional molecule is like using a square to stand for a cube, a triangle for a tetrahedron or prism, or a circle for a sphere, cone or cylinder. Each flat shape is a useful shorthand, but each loses the depth that defines the real object.

KEY POINT

One- and two-dimensional structures are enormously convenient, but they do not paint the true picture of molecular reality. Real drug molecules exist in three dimensions and interact with their protein receptors in three dimensions — so their spatial shape governs whether, and how well, they work.

§3 NATURAL PRODUCTS, STEREOCHEMISTRY AND THE PHARMACIST

A great many important drugs originate as natural products, and the chemistry of these substances — from their plant or microbial source through to their finished structure — is a field in its own right.

3.1 From source to structure

Natural-product chemistry follows a logical sequence: a promising compound is traced to its plant or microbial source, extracted, isolated in pure form, and then characterised structurally. Morphine, atropine, artemisinin, penicillin, erythromycin, hyoscine and vincristine are all familiar drugs whose stories begin in nature.

3.2 Stereochemistry and therapy optimisation

Because all drugs are chemical entities, their three-dimensional detail carries real therapeutic weight. A large proportion of drugs — on the order of a third to a half — contain stereocentres, display stereoisomerism and can exist as enantiomers, that is, as non-superimposable mirror-image forms. Since receptors are themselves three-dimensional and selective, two mirror-image forms of the same drug can differ markedly in activity, which is why stereochemistry is central to optimising therapy.

PRACTICAL RELEVANCE

Approach the subject where you stand, and do not be discouraged by claims that chemistry is too hard, too long or too dangerous to master. Get the recommended textbooks and read ahead, practise problems in abundance, attend every class and practical, form discussion groups, and pay attention to the smallest structural detail — it is often where a drug's behaviour is decided.

PART TWO

DRUGS IN THE LIQUID PHASE

§4 PHARMACEUTICAL SYSTEMS AND DISPERSIONS

The study of drugs in the liquid phase begins with the idea of a pharmaceutical system. A system is a bounded region of space, or a defined quantity of matter, that we choose to study. Material substances can be combined into a range of pharmaceutical mixtures, and classifying those mixtures is the first step towards understanding them.

4.1 Systems, phases and dispersions

Pharmaceutical mixtures, also called dispersions, fall into three broad classes: true solutions (molecular dispersions), colloidal dispersions, and coarse dispersions. A dispersion always contains at least two phases — one or more dispersed (internal) phases suspended within a single continuous (external) phase. A phase is a distinct, homogeneous part of a system that is set apart from other parts by a definite boundary. A true solution is a special case: a homogeneous molecular dispersion of two or more components forming a single phase.

4.2 Comparing the three dispersion types

Property	Molecular dispersion	Colloidal dispersion	Coarse dispersion
Particle size	Less than 1 nm	About 1 nm to 0.5 μm (500 nm)	Greater than 0.5 μm
Microscope visibility	Invisible even in the electron microscope	Visible in the electron microscope	Visible in an ordinary microscope
Passage through membranes	Passes semi-permeable membranes and ultra-filters	Passes filter paper but not semi-permeable membranes	Retained by ordinary filter paper and semi-permeable membranes
Example	Glucose in water (a true solution)	Jelly, butter, milk, shaving cream	Calamine lotion, fine sand in water, red blood cells (emulsions, suspensions)

§5 TRUE SOLUTIONS: SOLVENT, SOLUTE AND TYPES

Molecular dispersions — true solutions — are the main focus of liquid-phase pharmaceutical chemistry.

5.1 Binary solutions: naming the parts

A solution of just two substances is a binary solution, and its two constituents are the solvent and the solute. By convention the component present in the greater amount is called the solvent and the one present in the lesser amount the solute. There is one practical exception: when a solid is dissolved in a liquid, the liquid is taken as the solvent and the solid as the solute regardless of their relative amounts.

5.2 The nine types of solution

Because matter exists in three states — gas, liquid and solid — and a solute in any state may dissolve in a solvent in any state, nine combinations of homogeneous solute-solvent mixture are possible:

Solute	Solvent	Example
Gas	Gas	Air
Liquid	Gas	Water vapour in oxygen
Solid	Gas	Iodine vapour in air
Gas	Liquid	Carbonated water
Liquid	Liquid	Alcohol in water
Solid	Liquid	Aqueous sodium chloride solution
Gas	Solid	Hydrogen in palladium
Liquid	Solid	Mineral oil in paraffin
Solid	Solid	Gold-silver alloy; mixtures of alums

§6 PHYSICAL PROPERTIES OF SUBSTANCES

The physical properties of substances in solution can be sorted in two complementary ways: by how they depend on molecular identity, and by how they depend on the amount of matter present.

6.1 Colligative, additive and constitutive properties

- **Colligative properties** depend mainly on the number of solute particles present, not on their chemical identity. They include osmotic pressure, lowering of vapour pressure, depression of freezing point and elevation of boiling point. For equal concentrations of different non-electrolytes, these properties come out roughly the same regardless of the chemical nature of the solute.
- **Additive properties** are the sum of the contributions of the individual atoms or components. Molecular weight is the classic example, being the sum of the masses of the constituent atoms; likewise the total mass of a solution is the sum of the masses of its parts.
- **Constitutive properties** depend on the arrangement of atoms within a molecule, and only to a lesser extent on their number and kind. They give clues to how a compound is put together. Many real properties — refraction of light, electrical behaviour, surface and interfacial characteristics, and drug solubility — are partly additive and partly constitutive.

6.2 Extensive versus intensive properties

Class	Dependence on quantity	Examples
Extensive	Depend on the amount of matter in the system	Mass, volume
Intensive	Independent of the amount of substance present	Temperature, pressure, density, surface tension, viscosity of a pure liquid

§7 ELECTROLYTES, IDEAL GASES AND IDEAL SOLUTIONS

7.1 Classifying solutes

Solutes, whatever their physical state, divide into two broad classes. Non-electrolytes — such as sucrose, glycerine, naphthalene and urea — dissolve without forming ions and do not conduct electricity. Electrolytes — such as hydrochloric acid, sodium sulphate, ephedrine and phenobarbital — form ions in solution. Electrolytes are themselves subdivided:

- **Strong electrolytes** ionise essentially completely in water, for example hydrochloric acid (HCl) and sodium sulphate (Na_2SO_4).
- **Weak electrolytes** ionise only partially, for example ephedrine and phenobarbital.

7.2 Ideal gases and ideal solutions

An ideal gas is one in which there is no attraction between the molecules; it is a useful reference standard, and the behaviour of real gases approaches it as the pressure falls towards zero. By analogy, an ideal solution is one in which mixing the components produces no change in their properties other than the effect of dilution. Real gases and real solutions are understood by measuring how far they depart from these idealised limits.

KEY POINT

Colligative properties count particles, additive properties sum contributions, and constitutive properties reflect molecular arrangement. Whether a solute is a strong electrolyte, weak electrolyte or non-electrolyte determines how many particles it releases — and therefore how strongly it influences colligative behaviour.

PART THREE

BEHAVIOUR OF DRUGS IN BIOLOGICAL SYSTEMS**§8 WHAT ARE DRUGS? DEFINITION AND THE NEED FOR NEW DRUGS**

The central aim of medicinal chemistry is to design and discover new compounds fit for use as drugs. Doing so requires not only designing and synthesising a candidate molecule but also developing the tests and procedures needed to establish how it behaves in the body and whether it is suitable as a medicine.

8.1 Defining a drug

Following the World Health Organization, a drug is a chemical substance or product used, or intended to be used, to modify or explore a physiological system or a pathological state for the benefit of the recipient. A drug's activity is its pharmacological effect on the subject — for example an analgesic or beta-blocker action. Because drugs act by interfering with biological processes, no drug is entirely safe: even paracetamol, in overdose, can cause coma and death.

8.2 Beneficial and non-beneficial effects

Most drugs carry non-beneficial effects alongside their intended ones. Aspirin relieves headache but may irritate the stomach and cause bleeding. For some substances, such as cocaine and heroin, the undesirable effects are so serious that their use is tightly controlled by law. These unwanted effects are collectively called side effects.

8.3 Why new drugs are needed

- To combat the emergence of drug resistance.
- To improve the treatment of diseases for which therapy already exists.
- To treat newly identified diseases.
- To produce safer medicines by reducing or removing adverse drug reactions (ADRs).

§9 DRUG DISCOVERY AND DESIGN: A HISTORICAL OUTLINE

Humanity has drawn on natural products for medicine since ancient times. Remedies from animal, vegetable and mineral sources were sometimes strikingly effective, though many were also highly toxic. Modern practice calls the compound on which development is based a lead compound, and the related synthetic compounds derived from it its analogues. The odds remain long: it is estimated that of every 10,000 compounds synthesised, only about one proves suitable for medical use.

9.1 Ehrlich and the birth of rational synthesis

The first rational development of synthetic drugs is credited to Paul Ehrlich and Sacachiro Hata, who in 1910 produced the antiprotozoal arsphenamine (Salvarsan) by pairing chemical synthesis with reliable biological screening. Ehrlich recognised that both the beneficial and the toxic properties of a drug matter, and that the most effective drugs show greater selectivity for their target organisms than for the host. He expressed this as a chemotherapeutic index:

$$\text{Chemotherapeutic index} = \text{Minimum curative dose} / \text{Maximum tolerated dose}$$

Salvarsan was toxic but safer than the arsenical Atoxyl then in use; by the mid-1940s penicillin had superseded it. Ehrlich's index was later refined — to account for variation between individuals — into its reciprocal, the modern therapeutic index:

$$\text{Therapeutic index} = \text{Lethal dose } 50 \text{ (LD}_{50}\text{)} / \text{Effective dose } 50 \text{ (ED}_{50}\text{)}$$

9.2 From SAR to QSAR

Ehrlich's practice of synthesising and testing a series of structurally related compounds is now described as exploring the structure-activity relationship (SAR). In the 1960s Hansch and Fujita introduced quantitative measurement into this approach, creating QSAR (quantitative structure-activity relationships), which in the 1970s helped develop the antiulcer agents cimetidine and ranitidine.

9.3 Receptors, ligands and the pharmacophore

In 1905 John Langley proposed an alternative view of drug action: that receptive substances in the body can accept either a stimulating compound, which triggers a biological response, or a non-stimulating compound, which blocks one. It is now universally accepted that a chemical agent — a ligand — binding to a receptor sets off a chain of biochemical events producing a pharmacological effect. A drug works best when its molecular shape and electron distribution (its stereoelectronic structure) are complementary to those of the target receptor. The part of the ligand that actually binds is its pharmacophore. Side effects can arise when a drug binds not only to its intended receptor but also to others.

KEY POINT

The growth of understanding of disease chemistry and biological structure through the 20th century revealed how drugs distribute through the body, cross membranes, act and are metabolised — giving rise to the concept of pharmacokinetics.

§10 SOURCES OF DRUGS

Drugs are drawn from four broad origins: natural, synthetic, semi-synthetic and biosynthetic. Natural sources are themselves subdivided into plant, animal, microbiological and mineral origins.

10.1 Natural sources — plants

Plant source	Drug	Use / classification
Cinchona bark	Quinine	Antimalarial
Papaver somniferum (opium poppy fruit)	Morphine	Analgesic
Digitalis purpurea (purple foxglove)	Digoxin	Cardiac glycoside
Atropa belladonna (deadly nightshade)	Atropine	Anti-muscarinic agent
Artemisia annua (sweet wormwood)	Artemisinin	Antimalarial
Tobacco	Nicotine	Stimulant
Rauwolfia serpentina	Reserpine	Antihypertensive

10.2 Natural sources — animals, microbes and minerals

Animal source	Drug	Use / classification
Leech	Heparin	Anticoagulant

Animal source	Drug	Use / classification
Pork / cow pancreas	Insulin	Antidiabetic
Thyroid gland	Thyroxine	Hormone
Liver extract	Vitamin B12	Vitamin
Cow stomach	Pepsin	Digestive enzyme

Micro-organism	Drug
Penicillium notatum	Penicillin
Streptomyces venezuelae	Chloramphenicol
Penicillium griseofulvum	Griseofulvin
Streptomyces griseus	Streptomycin
Streptomyces fradiae	Neomycin

Mineral source	Drug	Use / classification
Kieserite, epsomite	Magnesium sulphate (MgSO_4)	Purgative / antacid, laxative
Melanterite, rozenite	Ferrous sulphate (FeSO_4)	Iron-deficiency anaemia
Nahcolite	Sodium bicarbonate (NaHCO_3)	Acidosis, indigestion / antacid
Gibbsite, bayerite	Aluminium hydroxide (Al(OH)_3)	Acidosis, indigestion / antacid
Zincite, zinc ores	Zinc oxide (ZnO)	Sunscreen / skin protectant
Calaverite, sylvanite	Gold (auranofin)	Rheumatoid arthritis / anti-inflammatory

10.3 Synthetic, semi-synthetic and biosynthetic sources

Synthetic drugs are compounds produced in the laboratory, often to mimic the effects of natural botanical compounds — examples include aspirin, oral antidiabetics, antihistamines, amphetamine, chloroquine and chlorpromazine. Their advantages are considerable: better quality control, an easier and cheaper process, the freedom to modify the prototype structure, products that can be safer and more potent, and suitability for large-scale production.

Semi-synthetic drugs are made by chemically altering a naturally obtained drug. This route is chosen when full synthesis of a complex molecule would be difficult, expensive or uneconomical, or when the natural source yields impure material. Examples include heroin from morphine, bromoscopolamine from scopolamine and homatropine from atropine (plant-derived), and the conversion of animal insulin to human insulin, and 6-aminopenicillanic acid derivatives (animal-derived).

Biosynthetic drugs are produced with the aid of biotechnology and genetic engineering, blending molecular biology, recombinant DNA (rDNA) technology, gene splicing and immunology. Such products are known as biologics or biopharmaceuticals; examples include recombinant human insulin, recombinant hepatitis B vaccine, menotrophin, urokinase and human growth-hormone-releasing hormone.

§11 ROUTES OF ADMINISTRATION AND DOSAGE FORMS

The physical form in which a medicine is given is its dosage form, which normally combines the active ingredient with other ingredients called excipients. Drugs may be given topically or

systemically, and the routes are classified broadly as either enteral (via the gastrointestinal tract) or parenteral (bypassing it, for example by injection).

11.1 Choosing a route

The route selected for a given drug depends on several factors working together:

- the physical and chemical properties of the drug;
- the site at which the drug is intended to act;
- the rate and extent to which the drug is absorbed from each possible route;
- the effect of digestive juices and of first-pass metabolism on the drug.

11.2 The journey through the body (overview)

Once given by an enteral route such as oral administration, a drug must cross the gastrointestinal membrane to be absorbed; any unabsorbed material is lost in the faeces. Absorbed drug enters the bloodstream and is carried to the liver, where first-pass metabolism may act on it, before continuing into the general circulation. From the blood the drug reaches its target site to produce the desired biological activity, may reach other sites to produce unwanted side effects, and is ultimately eliminated — for example through the kidneys in urine or the lungs in exhaled gases. A parenteral route such as intravenous injection places the drug directly into the bloodstream, bypassing gastrointestinal absorption and first-pass metabolism.

§12 PHARMACOKINETICS (ADME) AND PHARMACODYNAMICS

Two complementary frameworks describe what happens when a drug meets the body. Pharmacokinetics describes what the body does to the drug; pharmacodynamics describes what the drug does to the body.

12.1 The pharmacokinetic processes (ADME)

Process	What it involves
Absorption	Movement of the drug from its site of administration into the bloodstream
Distribution	Transport of the drug by the blood to the tissues and organs
Metabolism	Chemical alteration of the drug, mainly in the liver
Excretion	Removal of the drug through the kidneys, bile, lungs or sweat

12.2 Pharmacodynamics

Pharmacodynamics concerns the mechanism by which a drug produces its effect — typically through interaction with receptors, enzymes or ion channels. A drug that activates a receptor is an agonist; one that blocks it is an antagonist. The dose–response relationship shows how the effect changes with drug concentration, and the therapeutic window is the range of concentration lying between the effective and the toxic dose.

12.3 Integrating PK and PD

The two frameworks meet at the point of drug concentration: the concentration a drug reaches (governed by pharmacokinetics) determines the pharmacological effect it produces (governed by pharmacodynamics). Together they shape the onset, intensity and duration of a drug's action. Patient factors such as age, genetics and disease alter drug behaviour, which is why the same dose can affect two individuals quite differently.

PHARMACEUTICAL RELEVANCE

Understanding ADME explains why the same drug behaves differently by different routes: an oral dose must survive gastrointestinal absorption and first-pass metabolism, whereas an intravenous dose enters the circulation directly and completely. This is the chemical basis of rational prescribing.

PART FOUR

IONIC EQUILIBRIA

§13 CHEMICAL EQUILIBRIUM AND THE EQUILIBRIUM CONSTANT

Many reactions do not run to completion but settle into a balance between forward and reverse change. Understanding this balance is the foundation of everything that follows on pH and buffers.

13.1 Dynamic equilibrium

Consider a general reversible reaction in which a moles of A and b moles of B react to give c moles of C and d moles of D:



Let k_1 be the rate constant for the forward reaction (formation of C and D) and k_2 the rate constant for the reverse reaction (reformation of A and B). When the forward and reverse rates become equal, the reaction is in a state of dynamic equilibrium (chemical equilibrium). The forward rate is $v_1 = k_1[A]^a[B]^b$ and the reverse rate is $v_2 = k_2[C]^c[D]^d$. At equilibrium $v_1 = v_2$, so:

$$K = k_1/k_2 = [C]^c[D]^d / [A]^a[B]^b$$

Here each square-bracketed term is the concentration of that species raised to the power of its stoichiometric coefficient. The ratio K is the equilibrium constant, and its value fixes the position of the balance between reactants and products.

§14 WATER EQUILIBRIUM AND THE PH SCALE

Pure water is a very weak electrolyte that ionises to a tiny extent, and this self-ionisation defines the whole pH scale.

14.1 The ionic product of water, K_w

Water ionises according to $H_2O \rightleftharpoons H^+ + OH^-$. Applying the equilibrium condition and recognising that the concentration of water in dilute aqueous solution is effectively constant (about 55 mol/L), the water term can be folded into the constant to give the ionic product of water:

$$K_w = [H^+][OH^-] = 1.0 \times 10^{-14} \text{ (at } 25^\circ\text{C)}$$

In pure water every H^+ ion is matched by an OH^- ion, so if x mol/L of each is produced then $x^2 = 1.0 \times 10^{-14}$, giving $x = 1.0 \times 10^{-7}$ mol/L. Adding an acid raises $[H^+]$ above this value, and since the product must stay at 1.0×10^{-14} , $[OH^-]$ falls correspondingly; adding a base does the reverse.

14.2 Defining pH and pOH

The pH of a solution measures its hydrogen-ion concentration and runs from 0 to 14:

$$pH = -\log[H^+] \text{ and } pOH = -\log[OH^-]$$

Pure water, with $[H^+] = 1.0 \times 10^{-7}$ mol/L, has a pH of 7; all neutral solutions share this value. Acidic solutions have pH below 7 and basic solutions have pH above 7. Taking logarithms of the ionic-product expression gives the important companion relationship:

$$pH + pOH = 14$$

§15 DISSOCIATION OF WEAK ACIDS AND BASES

Weak acids and bases ionise only partially, and the extent of that ionisation is captured by their dissociation constants.

15.1 The Brønsted-Lowry picture

On the Brønsted-Lowry definition an acid is a proton donor and a base is a proton acceptor. An acid HA donates a proton to form its conjugate base A^- , while a base B accepts a proton to form its conjugate acid BH^+ . For acetic acid, $CH_3COOH \rightleftharpoons H^+ + CH_3COO^-$, the acid and its acetate ion form a conjugate acid-base pair. It is this incomplete dissociation that distinguishes weak acids and bases from strong ones such as HCl, HNO_3 and NaOH.

15.2 K_a , K_b and pK_a

The equilibrium constants for these partial dissociations are the dissociation constants K_a (for acids) and K_b (for bases). For the weak acid HA:

$$K_a = [H^+][A^-] / [HA] \quad pK_a = -\log_{10} K_a$$

A high K_a means the equilibrium lies well to the right, the acid is almost fully ionised, and it is strongly acidic; a low K_a means the equilibrium lies to the left and the acid is only weakly acidic. Because K_a values are often very small, the more convenient pK_a is used to compare acid strengths: the less positive the pK_a , the stronger the acid. For a conjugate acid-base pair the two constants are linked through water:

$$K_a \times K_b = K_w = 1.0 \times 10^{-14}$$

KEY POINT

For trichloroacetic acid (pK_a 0.7) versus acetic acid (pK_a 4.74), the smaller pK_a belongs to trichloroacetic acid — it dissociates more completely and is therefore the stronger acid. Comparing pK_a values is the quickest way to rank acid strength.

§16 THE HENDERSON-HASSELBALCH EQUATION

Relating the pH of a solution to the ratio of a weak acid and its salt gives one of the most useful working equations in pharmacy.

16.1 Deriving the equation

Starting from the acid dissociation expression $K_a = [H^+][A^-] / [HA]$ and taking negative logarithms of both sides, the hydrogen-ion term becomes pH and the constant becomes pK_a , leaving the Henderson-Hasselbalch equation for an acidic buffer:

$$pH = pK_a + \log ([salt] / [acid])$$

For a basic buffer, containing a weak base and its salt, the parallel forms are:

$$pOH = pK_b + \log ([salt] / [base])$$

$$pH = pK_a - \log ([salt] / [base])$$

16.2 Uses and limitations

- It calculates the pH of a buffer made from a mixture of a salt and a weak acid or base.
- It calculates the pK_a value from measured pH and known concentrations.
- It allows a buffer of a required pH to be prepared by fixing the salt-to-acid ratio.

The equation applies only to weak acids and bases in solution; it cannot be used for strong acids or strong bases, and it breaks down for acids with pK_a below about 2.0.

§17 BUFFER SOLUTIONS: MECHANISM AND TYPES

A buffer is a solution that resists changes in pH when small amounts of acid or base are added, neutralising them so that the pH stays relatively stable. This property, called buffer action, is vital wherever a process needs a specific, stable pH.

17.1 How a buffer works

A buffer contains a weak acid and its conjugate base in comparable amounts. Take a mixture of acetic acid and sodium acetate: the salt is fully ionised to acetate and sodium ions, so the solution holds both the weak acid CH_3COOH and its conjugate base CH_3COO^- . When a little acid is added, the extra H^+ is mopped up by acetate ions, which combine to form largely undissociated acetic acid: $\text{H}^+ + \text{CH}_3\text{COO}^- \rightleftharpoons \text{CH}_3\text{COOH}$. Because acetic acid is only slightly dissociated, it contributes almost no new H^+ , so the pH barely moves. When a little base is added, the OH^- is neutralised by the acetic acid: $\text{CH}_3\text{COOH} + \text{OH}^- \rightarrow \text{CH}_3\text{COO}^- + \text{H}_2\text{O}$, again leaving the pH almost unchanged. In each case the added acid or base is effectively absorbed and the pH is held nearly constant.

17.2 Characteristics and types

- A buffer has a definite pH value that does not change on standing for long periods.
- Its pH does not change appreciably on dilution.
- Its pH changes only very slightly when a small amount of a strong acid or base is added.

Type	pH range	Typical make-up	Example
Acidic buffer	Less than 7	A weak acid plus one of its salts (often a sodium salt)	Acetic acid + sodium acetate
Alkaline (basic) buffer	Greater than 7	A weak base plus one of its salts	Ammonia solution + ammonium chloride

The pH of a buffer can be tuned by changing the ratio of acid to salt, or by choosing a different acid-salt pair. Common combinations used in the laboratory include acetic acid / sodium acetate, phosphoric acid / potassium phosphate, oxalic acid / lithium oxalate, carbonic acid / sodium carbonate, and ammonia / ammonium chloride.

§18 BUFFERS IN PHARMACY AND BIOLOGICAL SYSTEMS

18.1 Why pharmaceutical products are buffered

The pH of pharmaceutical products is commonly adjusted with buffering systems for two main reasons: to secure maximum stability and to keep the pH within physiological limits. Pharmaceutical solutions are usually given a deliberately low buffer capacity so that they do not overwhelm the body's own buffer systems. A useful working rule is to choose a pharmaceutical buffer whose pH is greater than its pK_a . The main roles of buffers in pharmaceutical systems are:

- **Optimising biological activity** — enzymes work best over narrow pH ranges (pepsin, for example, is most active near pH 1.5), and the wrong pH can slow, impair or even permanently denature them.
- **Stability of compounds** — many drugs hydrolyse and become unstable in solution unless the pH is controlled; ascorbic acid and penicillin, for instance, are unstable in alkaline conditions.

Parenteral products are held near pH 7.4, the pH of blood, often using acetate, phosphate, citrate or glutamate buffers.

- **Solubility** — some ingredients dissolve only within a specific pH range, so buffering keeps the drug in solution.
- **Tissue compatibility** — isotonic preparations for delicate membranes are buffered to avoid irritation and tissue damage; an ideal parenteral pH is again 7.4.
- **Pharmaceutical analysis** — buffers of accurately known pH serve as standards for calibrating pH meters and preparing indicator colour standards.

18.2 Buffers in the living body

Blood is held at about pH 7.4 by several systems working together. Primary plasma buffers include carbonic acid / bicarbonate and the acid and alkali sodium salts of phosphoric acid; secondary buffers within the red cells include haemoglobin / oxyhaemoglobin and the potassium salts of phosphoric acid. Plasma proteins, behaving as acids, also combine with bases and act as buffers. It is generally life-threatening for blood pH to fall below 6.9 or rise above 7.8 — in diabetic coma it can drop to about 6.8. Tears show a notably high buffer capacity, tolerating a 1:15 dilution with neutral distilled water before their pH (about 7.4, range 7 to 8) shifts; urine has a pH near 6.0 (range 4.5 to 7.8), the kidney excreting hydrogen ions when pH falls and retaining them when it rises.

CLINICAL RELEVANCE

The tight buffering of blood at pH 7.4 is why parenteral products are formulated to match it: an injection that is strongly acidic or alkaline, or that carries too much buffer capacity, risks both local tissue damage and disturbance of the body's own acid-base balance.

§19 WORKED EXAMPLES AND TUTORIAL PROBLEMS

The following fully worked problems show how the equations of ionic equilibria are applied. Working through them by hand is the best way to make the ideas of pH, K_a and buffering secure before an assessment.

19.1 Finding pH from a strong acid or base

Example. Find the pH of (i) 0.10 M HCl and (ii) 0.10 M NaOH.

(i) HCl is a strong acid and dissociates completely, so $[H^+] = 0.10 \text{ M} = 1.0 \times 10^{-1} \text{ M}$, giving $\text{pH} = -\log(10^{-1}) = 1$.

(ii) NaOH is a strong base with $[OH^-] = 0.10 \text{ M}$; then $[H^+] = K_w / [OH^-] = 10^{-14} / 0.10 = 1.0 \times 10^{-13} \text{ M}$, so $\text{pH} = -\log(1.0 \times 10^{-13}) = 13$.

19.2 pH of a weak acid

Example. Calculate the pH of 0.01 M benzoic acid (C_6H_5COOH), with $K_a = 6.5 \times 10^{-5}$. Because equal amounts of $C_6H_5COO^-$ and H^+ form, $[H^+]^2 = K_a [C_6H_5COOH]$, so $[H^+] = \sqrt{(6.5 \times 10^{-5} \times 0.01)} = 8.06 \times 10^{-4} \text{ M}$, giving $\text{pH} = -\log(8.06 \times 10^{-4}) = 3.09$.

19.3 A buffer's pH by Henderson-Hasselbalch

Example. A buffer is 0.2 M in both acetic acid and sodium acetate; find its pH, then the pH after adding 10 mL of 1.0 M HCl to 500 mL of it. With $\text{p}K_a = -\log(1.8 \times 10^{-5}) = 4.74$ and equal salt and acid concentrations, $\text{pH} = 4.74 + \log(0.2/0.2) = 4.74$. Adding the acid supplies 0.01 mol H^+ in 0.5 L, i.e. 0.02 M, which converts 0.02 M of acetate into acetic acid: the acid rises to 0.22 M and the

salt falls to 0.18 M, so $\text{pH} = 4.74 + \log(0.18/0.22) = 4.66$ — a change of less than 0.1 pH unit, demonstrating the buffer at work.

19.4 Mass of salt needed to fix a buffer's pH

Example. What mass of sodium acetate must be added to 1.00 L of 0.2 M acetic acid (pK_a 4.76) to give pH 4.0? From $\text{pH} = \text{pK}_a + \log([\text{salt}]/[\text{acid}])$, $4.0 = 4.76 + \log([\text{salt}]/0.2)$, so $\log([\text{salt}]/0.2) = -0.76$ and $[\text{salt}] = \text{antilog}(-0.76) \times 0.2 = 0.035$ M. In 1.00 L that is 0.035 mol; with the molar mass of sodium acetate (CH_3COONa) = 82 g/mol, the required mass is $0.035 \times 82 = 2.87$ g.

19.5 Ratio of ionised to unionised drug at different pH

Example. A weakly acidic drug of pK_a 3.4 is present in two body compartments of pH 1.4 and 7.4. Using $\text{pH} = \text{pK}_a + \log([\text{ionised}]/[\text{unionised}])$: at pH 1.4, $\log([\text{ionised}]/[\text{unionised}]) = 1.4 - 3.4 = -2$, so the ratio is 1:100 (mostly unionised); at pH 7.4, the difference is $7.4 - 3.4 = 4$, so the ratio is 10,000:1 (almost entirely ionised). This is the chemical basis of pH-dependent drug absorption and distribution.

PRACTICAL RELEVANCE

Further tutorial problems worth practising include: the pH of 0.281 M $\text{Mg}(\text{OH})_2$; buffers prepared from potassium phosphate salts; the mass of ammonium chloride needed for a pH 9.10 ammonia buffer; and the mixing ratio of ammonia to ammonium chloride giving a chosen pOH. In every case, decide first whether the species is a strong or weak acid or base, then select K_w , the log definition of pH, or the Henderson-Hasselbalch equation accordingly.

PART FIVE

THERMODYNAMICS

§20 FOUNDATIONS OF THERMODYNAMICS

Thermodynamics is the branch of physical chemistry concerned with energy in all of its guises — how heat and work move between matter and its surroundings, and how substances respond when their temperature, pressure or volume is changed. Four fundamental laws (the Zeroth, First, Second and Third) frame the whole subject, capturing ideas such as thermal equilibrium, conservation of energy, the relentless growth of disorder, and the behaviour of matter as it approaches absolute zero. For a pharmacist these are not abstractions: they decide whether a formulation stays stable on the shelf, how readily a drug dissolves, and how tightly it will bind its biological target.

20.1 Why thermodynamics matters in pharmacy

Almost every stage of a medicine's life is governed by energy changes, so thermodynamic reasoning threads through formulation science, storage and drug action alike.

Application area	How thermodynamic thinking is used
Drug stability & shelf life	Predicts how temperature and humidity accelerate degradation, so that optimal storage conditions can be chosen to prolong shelf life.
Solubility & dissolution	Enthalpy and entropy of solution govern how well a drug dissolves in body fluids — a direct determinant of bioavailability.
Absorption & distribution	Partitioning of a drug between compartments such as blood and tissue is a thermodynamic problem; understanding membrane interactions guides delivery systems like liposomes and nanoparticles.
Dosage-form design	Explains how excipients in tablets, suspensions and emulsions interact, helping build stable products with the desired release profile.
Protein & enzyme stability	For biologic drugs, thermodynamics gauges the stability of protein folds and the conditions that guard against denaturation and loss of efficacy.

20.2 System, surroundings and boundary

Any thermodynamic analysis begins by drawing a mental line around the portion of the universe under study. That region is the system; everything outside it that can influence it is the surroundings; and the interface separating them is the boundary. A boundary may be rigid or flexible, insulating or permeable, and its properties decide what can cross it. Systems are classified by exactly what they let pass.

System type	Matter exchange	Energy exchange	Everyday example
Isolated	No	No	A sealed, perfectly insulated vacuum flask
Closed	No	Yes	A capped bottle that can warm or cool but loses no water
Open	Yes	Yes	A pot of boiling water losing both steam and heat

20.3 Thermodynamic processes

- **Isothermal** — carried out at constant temperature.
- **Adiabatic** — no heat crosses the boundary.
- **Isobaric** — carried out at constant pressure.

20.4 State functions

A state function is a property whose value depends only on the present condition of the system, never on the route taken to reach it. Because they are path-independent, state functions let us compare initial and final states directly and predict energy changes and equilibria with confidence.

State function	Symbol	Physical meaning
Internal energy	U	Total kinetic plus potential energy of all molecules in the system.
Enthalpy	H	Heat content at constant pressure; convenient for tracking heat in reactions.
Entropy	S	A measure of molecular disorder; points to the direction of spontaneous change.
Gibbs free energy	G	Combines H and S to predict spontaneity at constant T and P.

KEY POINT

Because state functions ignore the pathway, the enthalpy or entropy change for a reaction is fixed by the reactants and products alone — a fact that underlies both Hess's law (Section 27) and every free-energy calculation later in this guide.

§21 THE ZEROth LAW AND THERMAL EQUILIBRIUM

The Zeroth Law states that if two systems are each in thermal equilibrium with a third, then they are in thermal equilibrium with one another. Simple as it sounds, this is the statement that makes temperature a meaningful, measurable quantity: it establishes temperature as a transitive property.

21.1 What thermal equilibrium means

Two bodies are in thermal equilibrium when, placed in contact, there is no net flow of heat between them — they sit at the same temperature. If they start at different temperatures, heat always travels from the hotter to the cooler body until the two temperatures meet, at which point the flow stops. Drop a hot metal rod into cold water and heat leaves the rod for the water until both settle at one common temperature.

21.2 The thermometer as a go-between

The Zeroth Law is what lets a thermometer work. Suppose a thermometer (system C) is left in a hot cup of coffee (system A) until its reading steadies — the two are now in equilibrium. Move the same thermometer into a cup of water (system B) and again wait for a steady reading. By the Zeroth Law, coffee and water must be at the same temperature, even though they were never brought together. The identical logic lets a clinical thermometer under the tongue report body temperature once its reading stops changing.

PHARMACEUTICAL RELEVANCE

Every calibrated temperature scale — Celsius, Fahrenheit, Kelvin — rests on the Zeroth Law. Fixing 0 °C at ice-water equilibrium and 100 °C at boiling water gives reproducible reference points, which is why storage temperatures on a drug label mean the same thing in every pharmacy.

§22 THE FIRST LAW OF THERMODYNAMICS

The First Law is the law of conservation of energy: energy can be neither created nor destroyed, only converted from one form to another or shifted between system and surroundings. In a closed system the total energy is therefore constant, even as it changes guise — heat becoming work, for instance.

$$\Delta U = Q - W$$

Here ΔU is the change in the system's internal energy, Q the heat added to the system, and W the work the system does on its surroundings. Read plainly, the law says that energy fed into a system as heat must either raise its internal energy or be spent doing work — the accounting always balances.

22.1 Enthalpy and heat of reaction

Enthalpy (H) represents the total heat content of a system at constant pressure — loosely, the energy needed to create the system plus the energy needed to make room for it against its surroundings. When a reaction runs, the change in enthalpy ΔH tells us whether heat was released or absorbed, and reactions are classified accordingly.

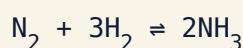
Reaction type	Sign of ΔH	Energy flow	Examples
Exothermic	$\Delta H < 0$	Heat released to surroundings	Combustion of wood or petrol; neutralisation of HCl by NaOH
Endothermic	$\Delta H > 0$	Heat absorbed from surroundings	Photosynthesis; dissolving NH_4NO_3 in water (the solution cools)

KEY POINT

Sign convention matters. A negative ΔH means the system loses heat (exothermic); a positive ΔH means it gains heat (endothermic). Getting the sign right is the difference between predicting a reaction that warms its container and one that chills it.

§23 REVERSIBILITY, EQUILIBRIUM AND K

A reversible reaction can run in both directions under suitable conditions. In a closed vessel the forward and reverse reactions proceed at the same time, and the synthesis of ammonia by the Haber process is the classic case.



Under high pressure and moderate temperature ammonia forms, yet it can also break back down into nitrogen and hydrogen if the conditions shift.

23.1 Chemical equilibrium and the equilibrium constant

Chemical equilibrium is reached when the forward and reverse rates become equal. The concentrations of reactants and products then hold steady — not because the reactions have stopped, but because they now cancel. The position of that balance is captured by the equilibrium constant K , the ratio of product to reactant concentrations, each raised to its stoichiometric coefficient.

$$K = [\text{Product}]^{\text{coefficient}} / [\text{Reactant}]^{\text{coefficient}}$$

- $K > 1$ — products are favoured at equilibrium.
- $K < 1$ — reactants are favoured at equilibrium.

For the gas-phase reaction $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$, the constant takes the form

$$K = [\text{HI}]^2 / ([\text{H}_2][\text{I}_2])$$

23.2 Le Chatelier's principle

Enthalpy also shapes how equilibrium responds to change. For an exothermic reaction, raising the temperature pushes the balance back towards reactants; for an endothermic reaction, the same heating drives it towards products. This is one face of Le Chatelier's principle: a system at equilibrium adjusts so as to oppose any change imposed on it, whether in temperature, pressure or concentration. A firm grasp of enthalpy, reversibility and K is what lets chemists steer industrial synthesis, pharmaceutical manufacture and energy production.

§24 THE SECOND LAW AND ENTROPY

The Second Law introduces entropy — a measure of disorder or the dispersal of energy — and asserts that the total entropy of an isolated system can only increase over time. The consequences are far-reaching: natural processes are irreversible, energy tends to spread out, heat flows spontaneously from hot to cold but never the reverse, and no heat engine can turn heat entirely into work without wasting some. This one-way character is often called the arrow of time.

24.1 Spontaneous processes

A spontaneous (or natural) process is one that proceeds of its own accord, with no outside help. Its reverse, which will not happen unaided, is non-spontaneous. Jumping off a cliff is spontaneous; climbing back up demands work. Heat passing from a hot metal ball to a cold one is spontaneous; forcing it the other way needs a refrigerator pump. A gas rushing to fill an evacuated vessel is spontaneous; compressing it back requires work.

Criteria of a spontaneous change

- It is unidirectional — reversing it requires work to be done on the system.
- Time is not a criterion — a spontaneous change may be fast or exceedingly slow.
- It continues as long as the system is away from equilibrium, and ceases once equilibrium is reached.
- An equilibrium system stays put unless external work disturbs it.
- It need not be exothermic: melting ice and evaporating water are endothermic yet spontaneous.

24.2 Spontaneity, randomness and entropy

For a long time chemists assumed only exothermic changes could be spontaneous, but melting ice on a warm day disproves that. What melting ice and evaporating water share is a rise in molecular randomness: the rigidly ordered crystal lattice of ice gives way to freely moving liquid molecules, which become freer still as vapour spreads through the air. A change that increases randomness is more likely to occur than one that imposes order — this new driving force is the entropy change, ΔS .

Entropy is a state function, so its change depends only on start and finish:

$$\Delta S = S_{\text{final}} - S_{\text{initial}}$$

When S_{final} exceeds S_{initial} , ΔS is positive and the process tends to be spontaneous. Statistically this makes sense: a highly disordered arrangement is far more probable than a highly ordered one, so systems drift naturally towards disorder.

24.3 Stating the Second Law

Applied to the universe — system plus surroundings — the law can be written in three equivalent ways:

$$\Delta S_{\text{univ}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}}$$

- For any spontaneous process, the entropy of the universe increases.
- For any spontaneous process, $\Delta S_{\text{universe}} > 0$.
- For any spontaneous process, $\Delta S_{\text{system}} + \Delta S_{\text{surroundings}} > 0$.

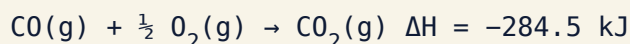
A reversible process leaves the universe's entropy unchanged ($\Delta S_{\text{univ}} = 0$), whereas an irreversible, spontaneous one raises it ($\Delta S_{\text{univ}} > 0$).

CAUTION

The Second Law does not require ΔS_{system} to be positive. The entropy of the reaction alone may fall; what the law demands is that the total — system plus surroundings — rises. If ΔS_{system} is negative for a spontaneous reaction, then $\Delta S_{\text{surroundings}}$ must be positive and large enough that their sum stays above zero.

§25 HEAT OF REACTION AND KIRCHHOFF'S EQUATIONS

The heat of reaction is the quantity of heat absorbed or released when the number of moles set out in a balanced equation reacts completely to products. For example, one mole of carbon monoxide burning with half a mole of oxygen to give carbon dioxide evolves 284.5 kJ:



By convention the reference or standard state is fixed at 298 K and one atmosphere. A heat change measured under those conditions is the standard enthalpy change, written ΔH° .

25.1 Variation of heat of reaction with temperature

Because a gas's specific heat changes with temperature, so does its heat of reaction. The equations describing this dependence are Kirchhoff's equations. At constant volume the heat of reaction ΔE is the difference between the internal energies of products and reactants; differentiating with respect to temperature and integrating between T_1 and T_2 gives:

$$\Delta E_2 - \Delta E_1 = \Delta C_v(T_2 - T_1)$$

where ΔC_v is the difference in heat capacities of products and reactants at constant volume. The parallel treatment at constant pressure, using enthalpy H , yields:

$$\Delta H_2 - \Delta H_1 = \Delta C_p(T_2 - T_1)$$

In words: the change in heat of reaction per degree is equal to the difference in heat capacities of products and reactants (at constant volume or constant pressure as appropriate). These relations together are Kirchhoff's equations.

WORKED EXAMPLE

Kirchhoff's equation. For $\frac{1}{2} \text{H}_2 + \frac{1}{2} \text{Cl}_2 \rightarrow \text{HCl}$ the heat of reaction at 27 °C is -22.1 kcal. The molar heat capacities (C_p) at 27 °C are 6.82 (H_2), 7.70 (Cl_2) and 6.80 cal mol⁻¹ (HCl). Find the heat of reaction at 77 °C.

$$\Delta C_p = 6.80 - \frac{1}{2}(6.82) - \frac{1}{2}(7.70) = 6.80 - 7.26 = -0.46 \text{ cal K}^{-1}.$$

$$\Delta H_2 = \Delta H_1 + \Delta C_p(T_2 - T_1) = -22\,100 + (-0.46)(50) = -22\,123 \text{ cal} \approx -22.12 \text{ kcal}.$$

§26 TYPES OF HEAT OF REACTION

The enthalpy change of a reaction is given a specific name according to the kind of process involved. The table below summarises the main types, each defined per mole of the substance concerned, followed by representative equations.

Type (symbol)	Definition — enthalpy change when...
Formation (ΔH_f)	one mole of a compound forms from its elements. Standard heat of formation uses elements in their standard states (298 K, 1 atm).
Combustion (ΔH_c)	one mole of a substance is completely burnt in excess oxygen. Always negative (heat is evolved).
Solution	one mole of substance dissolves in a solvent to the point where further dilution causes no further enthalpy change.
Neutralisation	one gram-equivalent of an acid is neutralised by one gram-equivalent of a base in dilute solution.
Fusion	one mole of a solid melts to liquid at its melting point.
Vaporisation	one mole of a liquid becomes vapour at its boiling point.
Sublimation	one mole of a solid passes directly to gas below its melting point.
Transition	one mole of an element changes from one allotropic form to another.

26.1 Standard heat of reaction from heats of formation

Standard heats of formation let us compute the standard heat of any reaction: it equals the summed standard heats of formation of the products minus those of the reactants.

$$\Delta H^\circ = \sum \Delta H_f^\circ (\text{products}) - \sum \Delta H_f^\circ (\text{reactants})$$

For a general reaction $aA + bB \rightarrow cC + dD$ this expands to

$$\Delta H^\circ = [c \cdot \Delta H_f^\circ(C) + d \cdot \Delta H_f^\circ(D)] - [a \cdot \Delta H_f^\circ(A) + b \cdot \Delta H_f^\circ(B)]$$

WORKED EXAMPLE

Standard heat of combustion of ethanol. Given ΔH°_f values of -277.0 ($\text{C}_2\text{H}_5\text{OH}$), -393.5 (CO_2) and $-285.5 \text{ kJ mol}^{-1}$ (H_2O), find ΔH° for $\text{C}_2\text{H}_5\text{OH(l)} + 3\text{O}_2\text{(g)} \rightarrow 2\text{CO}_2\text{(g)} + 3\text{H}_2\text{O(l)}$.

$$\Delta H^\circ = [2(-393.5) + 3(-285.5)] - [-277.0 + 0] = -1643.5 + 277.0 = -1366.5 \text{ kJ}.$$

(The standard heat of formation of the element O_2 is zero.)

PHARMACEUTICAL RELEVANCE

Heats of fusion and vaporisation are direct probes of intermolecular force: the larger the value, the stronger the attractions holding the substance together. This is why such data help explain melting points, boiling points and the solubility behaviour that governs drug formulation.

§27 HESS'S LAW OF CONSTANT HEAT SUMMATION

Hess's law states that if a chemical change can be brought about by more than one route — in a single step or through several — the total heat change is the same regardless of the path taken. It is a direct consequence of enthalpy being a state function.

If A converts directly to Z releasing Q_1 , while the same change carried out through intermediates B and C releases q_1 , q_2 and q_3 , then

$$Q_1 = q_1 + q_2 + q_3$$

Were this not so, one could cycle A to Z by one path and back by another and harvest energy endlessly from an isolated system — a clear violation of the First Law. Hence the heats must be equal.

WORKED EXAMPLE

Heat of formation of ethanol via Hess's law. Given (a) $\text{C}_2\text{H}_5\text{OH} + 3\text{O}_2 \rightarrow 2\text{CO}_2 + 3\text{H}_2\text{O}$, $\Delta H = -330.0 \text{ kcal}$; (b) $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$, $\Delta H = -94.3 \text{ kcal}$; (c) $\text{H}_2 + \frac{1}{2}\text{O}_2 \rightarrow \text{H}_2\text{O}$, $\Delta H = -68.5 \text{ kcal}$.

Take $2 \times (\text{b}) + 3 \times (\text{c})$: $2\text{C} + 3\text{H}_2 + 7/2 \text{O}_2 \rightarrow 2\text{CO}_2 + 3\text{H}_2\text{O}$, $\Delta H = -394.1 \text{ kcal}$.

Subtract (a): $2\text{C} + 3\text{H}_2 + \frac{1}{2}\text{O}_2 \rightarrow \text{C}_2\text{H}_5\text{OH}$, $\Delta H = -394.1 - (-330.0) = -64.1 \text{ kcal}$ — the heat of formation of ethanol.

§28 THE THIRD LAW OF THERMODYNAMICS

Entropy rises and falls with temperature. Follow water down the thermometer: as steam it is highly disordered, its molecules roaming the whole container; cooled to liquid they are confined but still mobile; frozen to ice they lock into an ordered crystal with little freedom. Cool the crystal further and lattice vibrations slow until, at absolute zero, all molecular motion ceases and the arrangement is perfectly ordered. This is the content of the Third Law:

KEY POINT

At absolute zero the entropy of a perfect pure crystal is exactly zero — that is, $S = 0$ at $T = 0 \text{ K}$. Formally, $\lim_{T \rightarrow 0} S = 0$.

Walther Nernst framed this principle, and because it enables the calculation of chemical affinity it is also called the Nernst heat theorem. Above 0 K the absolute entropy of a substance can be found by integrating the reversible heat exchange divided by temperature.

28.1 Importance and limitations

Importance	Limitations
Enables calculation of absolute thermodynamic properties.	Glassy (amorphous) solids retain some entropy even at 0 K.
Provides a route to chemical affinity (Nernst theorem).	Solids made of isotope mixtures do not reach zero entropy — e.g. solid chlorine.
Explains the behaviour of solids at very low temperature.	Crystals with residual disorder — CO, N ₂ O, NO, H ₂ O — keep non-zero entropy at 0 K.
Aids analysis of chemical and phase equilibria.	

§29 STANDARD AND NUMERICAL ENTROPY

In 1850 Clausius gave entropy a numerical footing. For a reversible change at a fixed temperature T , the entropy change equals the heat exchanged divided by that temperature:

$$\Delta S = q / T$$

Absorbing heat makes ΔS positive (entropy rises); releasing heat makes it negative. In SI units entropy is measured in $\text{J mol}^{-1} \text{K}^{-1}$.

29.1 Standard entropy

The Third Law fixes the entropy of a perfect crystal at zero at 0 K, so the actual (absolute) entropy of any substance at a higher temperature can be measured and calculated. The absolute entropy at 25 °C (298 K) and one atmosphere is the standard entropy, S° . Unlike enthalpies of formation, standard entropies of all substances above 0 K are positive. From tabulated S° values the standard entropy change of a reaction follows directly:

$$\Delta S^\circ = \sum S^\circ(\text{products}) - \sum S^\circ(\text{reactants})$$

WORKED EXAMPLE

Standard entropy change of a reaction. For the reduction $\text{Al}_2\text{O}_3(\text{s}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{Al}(\text{s}) + 3\text{H}_2\text{O}(\text{g})$, using S° values ($\text{J K}^{-1} \text{mol}^{-1}$) of 51 (Al_2O_3), 131 (H_2), 28 (Al) and 189 (H_2O):

$$\Delta S^\circ = [2(28) + 3(189)] - [51 + 3(131)] = 623 - 444 = +179 \text{ J K}^{-1}.$$

Entropy rises — gaseous water is more complex and disordered than hydrogen gas.

§30 GIBBS FREE ENERGY AND SPONTANEITY

Gibbs free energy (G) welds enthalpy and entropy into a single quantity that decides whether a reaction will proceed on its own at constant temperature and pressure. Its change is

$$\Delta G = \Delta H - T\Delta S$$

Sign of ΔG	Interpretation
$\Delta G < 0$	Spontaneous — the reaction proceeds forward without external energy; products are favoured.
$\Delta G > 0$	Non-spontaneous — the reaction needs an external energy input to go forward; reactants are favoured.
$\Delta G = 0$	Equilibrium — forward and reverse rates are equal and there is no net change.

30.1 Factors that influence spontaneity

- **Enthalpy** — exothermic reactions ($\Delta H < 0$) favour spontaneity.
- **Entropy** — an increase in disorder ($\Delta S > 0$) favours spontaneity.
- **Temperature** — raising T amplifies the $T\Delta S$ term, so high temperatures favour processes with positive ΔS ; low temperatures favour those driven by a negative ΔH .
- **Pressure** — higher pressure favours processes that reduce volume ($\Delta V < 0$).

A reaction that is both exothermic and entropy-increasing is spontaneous at all temperatures; one that is endothermic and entropy-decreasing is never spontaneous. The interesting cases are mixed, where temperature decides the outcome.

30.2 Temperature and pressure effects

The temperature dependence of G is governed by entropy: $(\partial G/\partial T)_P = -S$. Raising T makes the $-T\Delta S$ term more negative, favouring processes with $\Delta S > 0$ — melting of ice is favoured when warm, freezing of water when cold. The pressure dependence is governed by volume: $(\partial G/\partial P)_T = V$. Higher pressure favours volume-reducing changes such as a gas condensing to solid, while lower pressure favours volume-increasing ones such as vaporisation. Pressure effects are usually smaller than temperature effects because most reactions involve modest volume changes.

WORKED EXAMPLE

Is a reaction spontaneous? At 298 K, $\Delta H = 58.03$ kJ and $\Delta S = 176.6$ J K⁻¹. Find ΔG .

$\Delta G = \Delta H - T\Delta S = 58.03 - (298)(0.1766) = 58.03 - 52.63 = +5.40$ kJ — not spontaneous at this temperature.

WORKED EXAMPLE

Finding the crossover temperature. For $\text{Fe}_2\text{O}_3(\text{s}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{Fe}(\text{s}) + 3\text{H}_2\text{O}(\text{g})$, $\Delta H = +98.8$ kJ and $\Delta S = +141.5$ J K⁻¹.

At 298 K: $\Delta G = 98.8 - (298)(0.1415) = 98.8 - 42.2 = +56.6$ kJ — not spontaneous.

The reaction turns spontaneous once $T\Delta S$ exceeds ΔH , i.e. above $T = \Delta H/\Delta S = 98.8 / 0.1415 \approx 698$ K. Note the positive ΔH here refers to the system, not the surroundings.

§31 FREE ENERGY, EQUILIBRIUM AND K

The standard free-energy change ΔG° is the change when reactants in their standard states convert fully to products in their standard states. It cannot be measured directly, but the more negative it is, the further the reaction proceeds towards products before settling. Equilibrium is simply the point of lowest possible free energy.

31.1 Linking ΔG° to the equilibrium constant

At equilibrium the standard free-energy change and the equilibrium constant are tied together by one of the most useful relations in chemical thermodynamics:

$$\Delta G^\circ = -RT \ln K$$

where R is the gas constant (8.314 J K⁻¹ mol⁻¹) and T the absolute temperature. Away from equilibrium the free-energy change depends on the reaction quotient Q , which has the same form as K but uses current rather than equilibrium concentrations:

$$\Delta G = \Delta G^\circ + RT \ln Q$$

Combining these gives $\Delta G = RT \ln(Q/K)$, which tells us at a glance which way a reaction must move to reach balance.

Condition	Meaning for ΔG° and K	Meaning for ΔG and Q
Products favoured	$\Delta G^\circ < 0$, so $K > 1$	$\Delta G < 0$ ($Q < K$): reaction runs to the right
Reactants favoured	$\Delta G^\circ > 0$, so $K < 1$	$\Delta G > 0$ ($Q > K$): reaction runs to the left
At equilibrium	$\Delta G^\circ = 0$, so $K = 1$ (rare)	$\Delta G = 0$ ($Q = K$): no net change

KEY POINT

A more negative ΔG° means a larger K — a greater extent of product formation at equilibrium. This single equation lets thermodynamic data predict equilibrium constants, and measured equilibrium constants yield free energies in return.

§32 THERMODYNAMICS OF DRUG-RECEPTOR BINDING

Thermodynamics reaches its most directly pharmaceutical application in the way a drug binds its target. As Paul Ehrlich put it, a substance is not effective unless it is bound to another: a drug's action begins with its interaction with a receptor, and that interaction can be modelled as a simple binding equilibrium characterised by thermodynamic parameters — Gibbs energy, enthalpy, entropy and heat capacity.

32.1 Binding constants and free energy

The strength of a drug-receptor interaction is expressed through the association constant K_a or its reciprocal, the dissociation constant K_d :

$$K_a = [\text{complex}] / ([\text{drug}][\text{receptor}]) = 1 / K_d$$

The association constant connects directly to the Gibbs energy of binding, the governing potential at constant temperature and pressure:

$$\Delta G = -RT \ln K_a = 2.303 RT \log K_d = \Delta H - T\Delta S$$

The smaller K_d is, the greater the affinity of the drug for its receptor. That free energy splits into an enthalpic contribution (ΔH) and an entropic one ($T\Delta S$).

32.2 Enthalpy-entropy compensation

Drug-receptor binding is marked by enthalpy-entropy compensation: often one term favours binding while the other opposes it. Enthalpic contributions come from electrostatic forces, hydrogen bonds and van der Waals interactions; entropic contributions arise from several sources, and the loss of flexibility when a flexible ligand rigidifies on binding carries a real entropic cost.

Strategy to improve affinity	Thermodynamic rationale
Strengthen ligand-macromolecule contacts over ligand-solvent contacts	Yields a more favourable binding enthalpy (ΔH).
Make the ligand more hydrophobic	Improves the solvation entropy on binding (the hydrophobic effect).
Pre-shape the ligand to the binding-site geometry	Minimises the loss of conformational entropy upon binding.

PHARMACEUTICAL RELEVANCE

Thermodynamic data explain why a drug binds, not merely how strongly. Comparing ΔH and ΔS reveals the driving force — a large negative ΔH points to strong bond formation, a large positive ΔS to a dominant hydrophobic effect — and can help distinguish agonists from antagonists and guide the design of tighter-binding molecules.

PART SIX

CHEMICAL KINETICS

§33 REACTION KINETICS AND DRUG STABILITY

Both the synthesis of a drug and its eventual breakdown are chemical reactions, and both unfold at rates that reaction kinetics sets out to measure and explain. Because a drug product changes physically and chemically with time, kinetics is the tool that tells a pharmacist how quickly a medicine decomposes and how to slow that decay.

The period over which a product stays within specification under recommended storage is its shelf life, and stability must be assessed in the very container-closure system in which the drug is marketed. A working knowledge of degradation rates lets the pharmacist interpret conditions of instability, recommend stabilising measures, and advise patients — for instance by choosing a vehicle at the pH where a drug degrades most slowly, and avoiding harmful combinations.

33.1 Stability testing

Test	What it does
Real-time stability	Stores the product at the recommended conditions and monitors quality over its actual lifetime.
Accelerated stability (AST)	Speeds degradation with exaggerated conditions — raised temperature, humidity or light — to predict long-term behaviour and set a tentative expiry date.
Stress testing (ST)	Pushes beyond AST into severe conditions (e.g. 50–60 °C, ≥75% relative humidity) to probe how the product fails under extremes.

PHARMACEUTICAL RELEVANCE

Stability data set the shelf life and dictate storage advice. Aspirin, for example, is kept away from moisture because it degrades by hydrolysis — a direct, everyday application of reaction-rate thinking at the pharmacy counter.

§34 RATE OF REACTION, RATE LAWS AND ORDER

When reactants combine to form products, reactant concentrations fall with time while product concentrations rise. The rate of reaction is the change in concentration of a reactant or product over a time interval:

$$\text{rate} = \Delta C / \Delta t \text{ (instantaneously, } dC/dt)$$

34.1 The rate law and the law of mass action

For a general reaction $x\text{A} + y\text{B} \rightarrow c\text{C} + d\text{D}$, the law of mass action states that the rate is proportional to the product of the molar concentrations of the reactants, each raised to a power. The expression that captures this is the rate law:

$$\text{rate} = k[\text{A}]^x[\text{B}]^y$$

Here k is the rate constant, and the exponents x and y are the reaction orders with respect to each reactant. Both k and the orders must be found by experiment — they cannot be read off the

balanced equation.

34.2 Order of reaction

The order of a reaction is the number of concentration terms the rate depends on. The order with respect to one reactant is that reactant's exponent; the overall order is the sum of the exponents. For the alkaline hydrolysis of ethyl acetate,



the rate law is $\text{rate} = k[\text{CH}_3\text{COOC}_2\text{H}_5]^1[\text{NaOH}]^1$, so the reaction is first order in ester, first order in hydroxide, and second order overall.

34.3 Pseudo-order reactions

If one reactant is present in such large excess that its concentration barely changes, its contribution folds into the rate constant and the reaction behaves as if it were of lower order — a pseudo-order reaction. When aqueous NaOH and water swamp a little ethyl acetate, only the ester concentration changes appreciably, so $\text{rate} = k'[\text{ester}]$ with $k' = k[\text{NaOH}]$, and the reaction is pseudo-first-order.

WORKED EXAMPLE

Order of reaction (practice). For $(\text{CH}_3\text{CO})_2\text{O} + 2\text{C}_2\text{H}_5\text{OH} \rightarrow 2\text{CH}_3\text{CO}_2\text{C}_2\text{H}_5 + \text{H}_2\text{O}$, $\text{rate} = k[(\text{CH}_3\text{CO})_2\text{O}][\text{C}_2\text{H}_5\text{OH}]^2$. Order in acetic anhydride = 1; order in ethyl alcohol = 2; overall order = 3.

If the alcohol (the solvent) is in large excess, its concentration is effectively constant, so $\text{rate} = k'[(\text{CH}_3\text{CO})_2\text{O}]$ — a pseudo-first-order reaction.

§35 FIRST-ORDER REACTIONS

In a first-order reaction the rate depends on the concentration of a single reactant raised to the first power:

$$-d[A]/dt = k_1[A]$$

Integrating this rate law gives the working equations that connect concentration to time:

$$\ln[A] = \ln[A]_0 - kt \quad \log[A] = \log[A]_0 - kt/2.303$$

Reading the integrated equation

- It shows exactly how $[A]$ falls with time from its initial value $[A]_0$.
- A plot of $\log[A]$ against t is a straight line of slope $-k/2.303$ — and a straight line is the diagnostic test that a reaction really is first order in A.
- Given $[A]_0$ and k , the concentration at any later time can be calculated.

35.1 Half-life and shelf-life

The half-life $t_{1/2}$ is the time for a reactant to fall to half its initial value; the shelf-life t_{90} is the time for it to fall to 90% of its original amount (a 10% loss). Setting $[A] = [A]_0/2$ in the integrated law gives the characteristic first-order results:

$$t_{1/2} = 0.693 / k \quad t_{90} = 0.105 / k$$

KEY POINT

For a first-order reaction the half-life depends only on the rate constant k , never on the initial concentration. This concentration-independence is the hallmark of first-order kinetics and is why radioactive decay and many drug-elimination processes have a fixed half-life.

WORKED EXAMPLE

First-order half-life. A first-order reaction has a half-life of 20.0 min. Find k , and the time to reach 75% completion.

$$k = 0.693 / t_{1/2} = 0.693 / 20.0 = 0.0347 \text{ min}^{-1}.$$

At 75% complete, $[A]/[A]_0 = 0.25$, so $t = \ln(4)/k = 1.386 / 0.0347 \approx 40 \text{ min}$ (two half-lives, as expected).

WORKED EXAMPLE

Assigning an expiry date. A drug decomposing by first-order kinetics starts at 94 units mL^{-1} with $k = 2.09 \times 10^{-5} \text{ hr}^{-1}$ at 25 °C, and is unusable below 45 units mL^{-1} .

$t = \ln([A]_0/[A]) / k = \ln(94/45) / (2.09 \times 10^{-5}) = 0.7369 / 2.09 \times 10^{-5} \approx 35\,260 \text{ hr} \approx 4 \text{ years}$ — the expiry interval to assign.

§36 SECOND-ORDER REACTIONS

For a reaction that is second order in a single reactant, $aA \rightarrow \text{products}$, the rate law is

$$-d[A]/dt = k[A]^2$$

For a bimolecular reaction $A + B \rightarrow \text{products}$ in which the rate is first order in each reactant, the decompositions of A and B keep pace and the rate is proportional to the product of the two concentrations, $\text{rate} = k[A][B]$. Integrating the single-reactant case gives

$$1/[A] = kt + 1/[A]_0 \quad t_{1/2} = 1 / (k[A]_0)$$

Key features

- Doubling $[A]$ quadruples the rate, because rate goes as the square of concentration.
- A plot of $1/[A]$ against t is a straight line of slope k — the diagnostic test for second-order kinetics.
- Unlike the first-order case, the half-life does depend on the initial concentration $[A]_0$ as well as on k .
- With k and $[A]_0$ known, $[A]$ at any time follows from the integrated equation.

§37 ZERO-ORDER REACTIONS

A zero-order reaction is one whose rate does not depend on reactant concentration at all:

$$-d[A]/dt = k[A]^0 = k$$

The rate is therefore constant, and integrating gives a concentration that falls linearly with time:

$$[A]_t = [A]_0 - k_0 t$$

The half-life and shelf-life both depend on the initial concentration as well as on k :

$$t_{1/2} = [A]_0 / (2k_0) \quad t_{90} = 0.10[A]_0 / k_0$$

WORKED EXAMPLE

Zero-order concentration and time. A zero-order reactant starts at 0.75 M with $k = 0.015 \text{ M min}^{-1}$.

After 15 min: $[A] = 0.75 - (0.015)(15) = 0.75 - 0.225 = 0.525 \text{ M}$.

To fall to 0.06 M: $t = ([A]_0 - [A]) / k = (0.75 - 0.06) / 0.015 = 46 \text{ min}$.

37.1 Units of the rate constant

Because the rate law changes with order, so do the units of k . Reading the units of a reported rate constant is itself a quick check on the reaction order.

Order of reaction	Units of rate constant k
Zero order	$\text{mol L}^{-1} \text{ s}^{-1}$
First order	s^{-1} (or hr^{-1})
Second order	$\text{L mol}^{-1} \text{ s}^{-1}$

§38 FACTORS AFFECTING REACTION RATE

Several controllable variables raise or lower the speed of a reaction, and each has a clear molecular explanation rooted in how often, and how energetically, particles collide.

Factor	Effect on rate and why
Temperature	Higher temperature gives particles more energy and more frequent, more forceful collisions, so the rate rises.
Surface area	A larger surface on a solid reactant exposes more particles to collision, speeding the reaction.
Concentration	For first- and second-order reactions, raising reactant concentration increases the collision frequency and the rate.
Pressure	For reactions involving gases, higher pressure packs molecules closer, increasing collisions and rate.
Catalysis	A catalyst lowers the activation energy for the reaction, allowing more collisions to succeed and raising the rate.

§39 ENZYME KINETICS AND MICHAELIS-MENTEN

Catalysts speed reactions by lowering activation energy, and in biology the great catalysts are enzymes — large protein molecules tailored to accelerate a specific reaction. Enzyme kinetics is the study of enzyme-catalysed reactions. Acetylcholinesterase, for instance, breaks down the neurotransmitter acetylcholine into choline and acetic acid. Catalysts are either homogeneous (in the same phase as the reactants) or heterogeneous (in a different phase, typically a solid).

39.1 The enzyme-substrate mechanism

Catalysis proceeds through a transient complex. The enzyme (E) and substrate (S) first bind to form an enzyme-substrate complex (ES), the reaction occurs, and the enzyme then releases the product (P) and is immediately free to act again:



Because a single enzyme molecule turns over its reaction repeatedly and rapidly, only a tiny amount of enzyme is needed to process a great deal of substrate — the source of the enzyme's remarkable efficiency.

39.2 The Michaelis-Menten equation

Michaelis-Menten kinetics models how the rate of an enzyme-catalysed reaction depends on substrate concentration:

$$V = V_{\max} [S] / (K_m + [S])$$

Term	Meaning
V	Initial rate (velocity) of the enzymatic reaction.
$V_{\max}$	Maximum rate, reached when every active site is saturated with substrate.
[S]	Substrate concentration.
K_m (Michaelis constant)	The substrate concentration at which the rate is half of $V_{\max}$. It gauges enzyme-substrate affinity: a low K_m means high affinity, a high K_m low affinity.

KEY POINT

At low substrate levels the rate climbs almost linearly with [S]; as [S] rises the active sites fill and the curve bends over, approaching $V_{\max}$ asymptotically. K_m — the [S] giving half $V_{\max}$ — is a compact, clinically useful measure of how tightly an enzyme grips its substrate, and underpins the way many drugs are designed to inhibit specific enzymes.

Originality & Intellectual-Property Disclaimer

© EverythingABUAD | everythingabuad.com. All content, explanations, and practice questions in this document were written from scratch by the EverythingABUAD student team as an original study aid. This document is not an official examination paper, is not affiliated with or endorsed by any university or college, and does not predict any future test. Always cross-check with your lecturer's current course outline. Free to use, not for resale.

An original study guide by the EverythingABUAD team.

everythingabuad.com